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Car Price Prediction using Regularized Regression

Section1: Data Summary

- Dataset: Car sales dataset (Size:157x16)
- Target variable: Price of cars(Price_in_thousands)
- Feature variables:Car specks (Horsepower, Engine_size, Curb_weight, Fuel_efficiency, Wheelbase, Fuel_capacity)

Section2: Objective of the analysis

The goal of this project is to model and predict car prices using vehicle specks and understand which specifications most strongly influence them using linear regression techniques. By comparing ordinary least squares(OLS) against regularized methods (L1,L2) and polynomial features through cross validation, the analysis aims to determine the most accurate and statistically stable model.

Section3: Model Comparison

To predict car prices and examine linear relationships, we formulated five regression setups: baseline Ordinary Least Squares (OLS), Ridge (L2 regularization), Lasso (L1 regularization), and second-degree Polynomial Features combined with both Ridge and Lasso. All models were built inside scikit-learn pipelines with standard feature scaling to prevent data leakage during preprocessing. For Ridge and Lasso variants, regularization penalties (α) were tuned dynamically using internal cross-validation across logarithmic grids. Final model performance was evaluated using 5fold cross validation, recording the R2 score and Root Mean Squared Error (RMSE) to identify the best balance between complexity and predictive accuracy.

Section4: Key Findings

	Model	Best Alpha	CV R2	CV RMSE ($k)
0	OLS (Baseline)	-	0.714675	7.676474
1	Ridge	1.0	0.718870	7.619832
2	Lasso (L1)	0.0001	0.708259	7.762297
3	PF + Ridge	10.0	0.693013	7.962543
4	PF + Lasso (L1)	0.022122	0.667037	8.292589

As shown in the comparison table, Ridge regression achieved the strongest overall performance, outperforming both the baseline OLS and the polynomial variants. Inspecting the standardized coefficients of the winning Ridge model reveals that Horsepower is by far the most influential positive driver of vehicle price (β≈+14.26), followed by Curb weight (β≈+4.59). Conversely, Engine size exhibits a negative coefficient (β≈−4.69β≈−4.69), which reflects multicollinearity with horsepower indicating that once engine output is accounted for, larger displacement does not independently command a premium.The intercept at 27.33 represents the baseline average price for an average specified vehicle.

Section5: Limitations and next steps

A key limitation of this project is the modest sample size (approximately 150 instances) and the presence of right-skewed pricing outliers, which constrained more complex polynomial architectures and induced mild overfitting. Future work should explore a logarithmic transformation on vehicle prices log(y) to stabilize variance, expand data collection across broader manufacturing segments, and benchmark against tree-based ensembles (such as Random Forests or Gradient Boosting) to capture non-linear market interactions without manual feature engineering.